A Transformer-Based Framework with Dynamic Multiscale Attention Mechanisms for 12-Lead ECG Synthesis

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***Abstract*— Cardiovascular disease (CVD) remains the leading cause of mortality worldwide, emphasizing the need for enhanced diagnostic methods. Standard 12-lead Electrocardiograms (ECGs) are the gold standard for CVD diagnosis but require specialized equipment, limiting diagnostic accessibility. Wearable smart devices enable at-home heart monitoring but are only able to record a limited lead set, restricting their diagnostic scope. This paper proposes a methodology that uses only three leads as input—I, III, and V6—to generate a complete 12-lead ECG. The key innovation proposed in this paper is the Dynamic Multi-Scale Time Series Transformer (DMTT), which synthesizes leads V1-V5 using only the three input leads. ECG synthesis is challenging due to the presence of both sharp and subtle waveform variations. DMTT integrates Dynamic Window and Multi-Scale architectures, enabling precise synthesis of ECG morphology and timing. DMTT consists of two primary components: the Dynamic Window Module (DWM) and Multi-Scale Module (MSM). The DWM implements a saliency-based attention mechanism, which is a small sub-network that infers local relevance for each time step, generating a saliency map that enables the model to adaptively pool features based on their measured importance. The MSM pools the sequence at multiple temporal scales and integrates these representations, allowing the model to capture spatiotemporal dependencies more effectively. This mechanism facilitates accurate synthesis of ECGs. The model was trained on the PTB-XL dataset and was evaluated using two accuracy metrics: Normalized Root Mean Square Error (NRMSE) and Average Absolute Error (AAE). Results demonstrate that the synthesized ECGs achieve an average AAE of 0.0777mV and NRMSE of 0.0907 across leads V1-V5 and five disease types. Qualitative analysis confirms the preservation of diagnostic features, illustrating the feasibility of synthesizing a 12-lead ECG from a subset of leads using a generative AI model.**

***Clinical Relevance*— The novel technique of synthesizing a full 12-lead ECG from a limited lead set allows wearable devices to capture a subset of leads, which serve as inputs to DMTT for generating a complete 12-lead ECG. This approach facilitates at-home heart health monitoring for a broad range of heart diseases using wearable technology. Additionally, it enhances the applicability of automated AI classification methods, enabling the detection of cardiac conditions that manifest in any of the 12 ECG leads.**

I. Introduction

Cardiovascular disease (CVD) stands out as the leading cause of death worldwide for the past three decades, underscoring the need for effective diagnostic tools [1]. Electrocardiograms (ECGs) serve as the gold standard for CVD detection. A standard ECG is made up of 12 leads, each recording the electrical activity of the heart from a different anatomical perspective [2]. Leads I, II, and III are classified as limb leads and are responsible for recording the heart's electrical activity within the frontal plane [3]. Leads aVR, aVL, and aVF, collectively known as the augmented limb leads, provide an enhanced perspective of the heart's electrical activity within the frontal plane by amplifying signals derived from different electrode pairings [4],[5]. The remaining six leads, V1 through V6, are referred to as precordial leads and record the heart's electrical activity in the horizontal plane. This combination of limb and precordial leads creates a three-dimensional view of the heart's electrical activity, enabling clinicians to detect abnormalities or diseases [6]. A single ECG waveform represents the electrical activity of an individual heartbeat. The P wave corresponds to atrial depolarization, the QRS complex signifies ventricular depolarization, and the T wave reflects ventricular repolarization. Fig. 1 shows a pictorial representation of a single heartbeat highlighting the various parts. The timing and morphology of the P, QRS, and T waves are key indicators of heart health. Abnormalities in any part of the ECG can be a clear indication of either past or current CVD [7]. Each disease usually has a unique morphology and timing associated with it; some examples are shown in Fig. 2. Capturing the subtle slopes of P and T waves without losing the sharp transitions of QRS complexes presents a significant challenge.

II. Methodology Overview

This paper proposes a three-part methodology to arrive at a full 12-lead ECG from a limited lead set as shown in Table 1.

1) Leads I, III, and V6 are chosen as the input using which all other leads are generated. These leads were chosen because

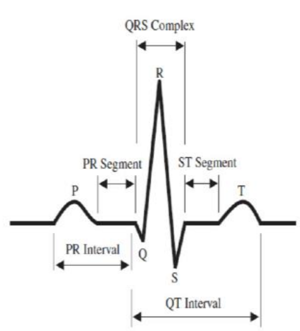

Figure 1. Parts of an ECG [7]

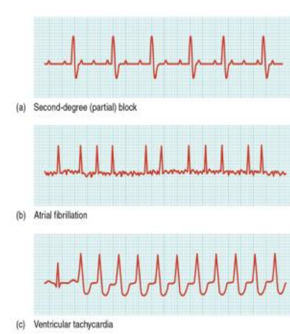

Figure 2. ECG abnormalities [8]

Table 1. Computation and Synthesis methods for arriving at a 12-lead ECG

Lead	**Synthesis Method**
I	Input Lead
III	Input Lead
V6	Input Lead
II	Einthoven's Lead System (1)
aVR	Goldberger's Lead System (2)
aVL	Goldberger's Lead System (3)
aVF	Goldberger's Lead System (4)
V1	Synthesize Using Neural Network
V2	Synthesize Using Neural Network
V3	Synthesize Using Neural Network
V4	Synthesize Using Neural Network
V5	Synthesize Using Neural Network

they capture the electrical activity of the heart across the three planes. They were also selected due to their compatibility with smart devices, such as the Apple Watch™ or Kardia™ [9], [10]. For this paper, all the designated input leads (i.e., I, III, and V6) are obtained from the PTB-XL database [11].

2) Einthoven's and Goldberger's formulae are utilized to compute Lead II, aVR, aVL, and aVF using only the input leads.

$$Lead\ II = Lead\ I + Lead\ III \tag{1}$$

$$aVL = \frac{Lead\ I - Lead\ III}{2} \tag{2}$$

$$-aVR = \frac{Lead\ I + Lead\ II}{2} \tag{3}$$

$$aVF = \frac{Lead\ II + Lead\ III}{2} \tag{4}$$

3) To synthesize the remaining five leads (i.e., V1-V5), a trained neural network is employed, leveraging the three input leads to generate accurate ECGs. This research paper focuses on the neural network design for synthesizing V1-V5.

III. Methods

A. Data Preparation

The 12-lead ECGs that were used in this study were sourced from the PTB-XL dataset, one of the largest publicly available ECG datasets with more than 21,799 ECGs from over 18,000 patients. ECGs that were documented to have static noise, burst noise, baseline drift, electrode problems and from patients with pacemakers were removed resulting in 15,190 ECGs. From the 15,190 ECGs, five cardiovascular conditions were selected: Normal, Inferior Myocardial Infarction (IMI), Anteroseptal Myocardial Infarction (ASMI), Left Ventricular Hypertrophy (LVH), and Left Anterior Fascicular Block (LAFB). To standardize the dataset, three individual beats were extracted from each ECG. However, due to an uneven distribution among four of the disease categories, additional beats were extracted to ensure that each condition contained a total of 4,500 ECG recordings. The extracted single-beat ECGs were subsequently processed using a high-pass filter to remove baseline drift and static noise. The ECGs were inspected to ensure that the high-pass filter was clinically insignificant. The final dataset then included a total of 22,500

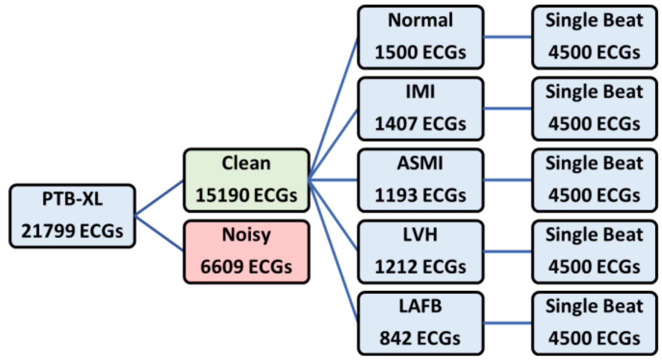

Figure 3. Data selection and preparation

individual beats, each split equally among the five disease types to minimize any bias. This final dataset of 22,500 one-beat ECGs were split 70-30, resulting in 15,750 ECG samples for training and 6,750 ECG samples for validation of the synthesis model.

B. Model Overview

Several types of generative AI models were explored and evaluated for this research including, Long Short-Term Memory Networks (LSTMs), Convolutional Neural Networks (CNNs), and Generative Adversarial Networks (GANs) [12]. Finally, the Transformer model was selected for its capacity to track both long and short-term patterns while leveraging attention mechanisms to capture fine morphological details in ECG signals [13]. This study proposes Dynamic Multi-Scale Time Series Transformer (DMTT), a novel transformer-based model that integrates a custom combination of, multi-head attention, multi-scale attention and dynamic attention windows. As shown in Fig. 4, DMTT consists of two parts: a Dynamic Window Module (DWM) and Multi-Scale Module (MSM), that work together to capture spatiotemporal dependencies across different resolutions and scales. Residual connections and layer normalization were used to stabilize training and improve gradient flow. A feedforward network (FFN) with ReLU activation and dropout was also applied to the output of the attention mechanisms.

C. Dynamic Saliency Based Attention Mechanism

This paper implements the Dynamic Window Module (DWM) using a saliency-based attention mechanism, which learns to highlight the most critical values in a time series by generating a saliency map $s \in R^{B\times T\times 1}$ from an input of $X \in R^{B\times T\times D}$. Unlike traditional methods that use sliding windows with fixed-length segmentation, this dynamic approach scales the window sizes based on the learned importance of values. This allows the windows to be updated dynamically, eliminating the boundaries posed by strict temporal windows.

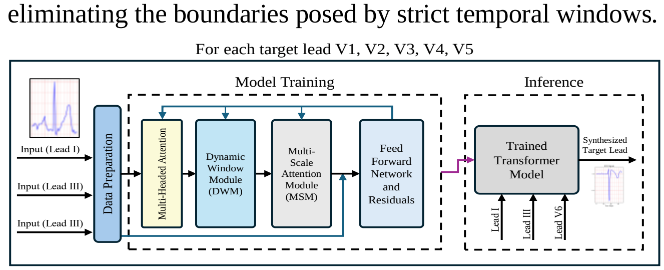

Figure 4. Dynamic Multi-Scale Time Series Transformer (DMTT) model architecture

Input X is passed through a small neural sub-network to produce intermediate representation:

$$Z_1 = ReLU(XW_1 + b_1) \tag{5}$$

$$Z_2 = ReLU(Z_1 W_2 + b_2) \tag{6}$$

Then, the saliency map s is obtained by

$$s = \sigma(Z_2 W_s + b_s) \tag{7}$$

where σ is the sigmoid function which ensures that each step's saliency is between [0,1]. Next, X, the original features are mapped into a new feature space.

$$F = XW_f + b_f \tag{8}$$

with W_f and b_f being trainable parameters. Parallelly, the saliency map s is expanded to yield window heights.

$$w = \mathrm{softmax}(sW_w + b_w) \tag{9}$$

This allows the various pooling windows to be emphasized according to the local saliency. Using these weights the network dynamically pools F at the various window sizes, sums the pooled representations, and reprojects them yielding a contextually refined feature map or F':

$$F_i' = Pool_i(F) \odot w_i \tag{10}$$

where $\odot$ represents element wise multiplication. The refined feature map is now ready for input into the multi scale module (MSM). The resultant refined feature map highlights the various morphological patterns in the ECG signal, becoming particularly useful for capturing smaller details in the ECG like the P-wave.

D. Multi-Scale Attention

The Multi-Scale Attention Module (MSM) implemented in this paper is designed to capture patterns at multiple temporal resolutions by incorporating simultaneous processing streams for different window sizes, then learning how to weigh these streams adaptively. By processing the refined feature map at multiple scales, the model can capture both fine-grained and coarse-grained patterns. Combining information from multiple scales leads to a richer and more informative representation of the input, which captures patterns within and across each of the distinct parts of the P, QRS, and T waves. The key parts of the MSM implementation are as follows.

The MSM receives the refined feature map from DWM as input. For a set of predefined scales s, the MSM applies scale specific poling or down sampling to F', creating

$$\tilde{F}^{(s)} = pool_s(F') \tag{11}$$

Each pooled tensor is then transformed by

$$Z^{(s)} = \tilde{F}^{(s)} W^{(s)} + b^{(s)} \tag{12}$$

Scale specific attention (SSA) calculations are then performed as follows

$$\alpha^{(s)} = \mathrm{softmax}\left(\tanh\left(Z^{(s)}\right) v^{(s)}\right) \tag{13}$$

which assigns an importance weight at each time step for a particular scale. A trainable parameter, $\gamma^{(s)}$, captures how globally relevant a scale s is, which is enforced to be positive by passing it through a Softplus activation. Thus, the attended representation at scale s is

$$A^{(s)} = Z^{(s)} \odot \alpha^{(s)} softplus\left(\gamma^{(s)}\right) \tag{14}$$

After computing $A^{(s)}$ for all scale, MSM concatenates them along the feature dimension into C, and finally projects them into a suitable output dimension with

$$Y = CW_{combine} + b_{combine} \tag{15}$$

By combining local saliency-driven pooling the refined feature map and multi-resolution attention, the architecture jointly exploits both fine-grained focus and global structure, enabling it to learn complex temporal dependencies with a focus on the most crucial segments. This enables not only accurate synthesis of P, QRS, and T waves but also in precise generation of ECG intervals, enhancing their diagnostic quality.

E. Residual and Feed Forward Layers

In addition to the Dynamic Window Module (DWM) and Multi-Scale Module (MSM), the model integrates feed-forward networks (FFNs), residual connections, and layer normalization. These components facilitate stable training and enhance the efficient extraction of features from the DWM and MSM. Residual connections were placed between the multi-head attention and the DWM, allowing the input to bypass the intermediate data transformations such as the MSM and DWM, helping alleviate the vanishing gradient problem. Layer normalization was applied after the MSM to accelerate training convergence. Feed forward networks (FFNs) were used to function as position-wise transformations, consisting

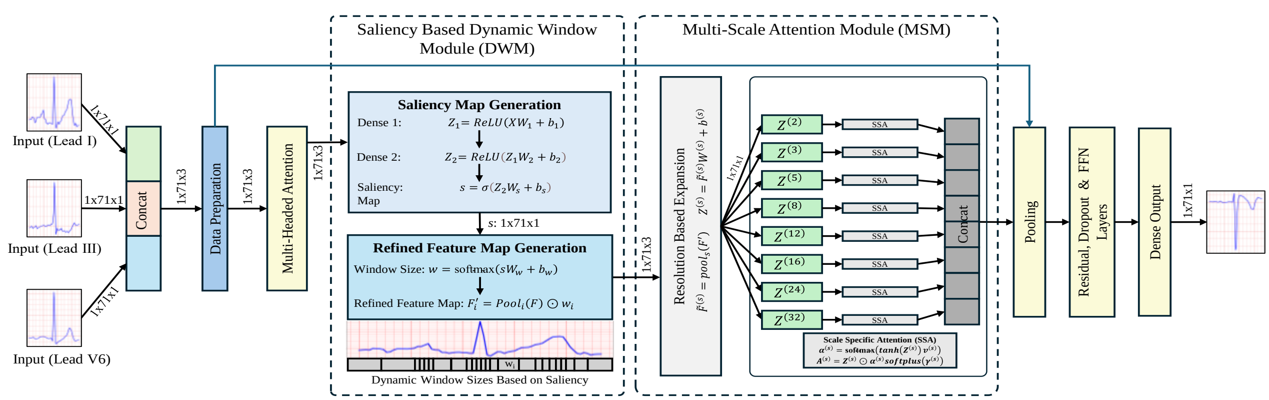

Figure 5. Architecture of Dynamic Window Module (DWM) and Multi-Scale Attention Module (MSM)

of two dense layers with a ReLU activation, expanding the dimensionality in the intermediate layer before projecting back down, helping the model capture complex patterns.

F. *Loss Function*

DMTT employs Symmetric Mean Absolute Percentage Error (SMAPE) as the loss function. Conventional loss metrics like Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) were unsuitable for synthesis of ECGs, as ECGs are composed of regions of sharp transitions near the QRS complex and small transitions at the P and T waves. RMSE and MSE disproportionally penalize large errors near areas of sharp transitions because MSE and RMSE loss functions both square the errors. Errors occurring in regions of sharp transitions, such as the QRS complex, are penalized more heavily due to the higher overall magnitude of the waveform in these areas, resulting in larger error values. This results in overfitting in areas of sharp transitions while performing poorly in areas with smaller transitions, or vice versa. SMAPE is a loss function specifically designed for time series data with a balanced sensitivity; it treats sharp and small transitions equally, ensuring that the model does not overemphasize large errors at the expense of small errors. SMAPE is also less affected by outliers, making it more robust. As shown in (16), SMAPE returns proportional errors rather than absolute deviations. This accounts for the scale of the values and ensures that errors in sharp and small transitions contribute meaningfully to the overall loss.

$$SMAPE = \frac{1}{N}\sum_{i=1}^{N}\frac{|s_{true,i}-s_{pred,i}|}{|s_{true,i}|+\frac{|s_{pred,i}|}{2}} \times 100 \qquad (16)$$

where $s_{true,i}$ denotes the true ECG signal while $s_{pred,i}$ represents the synthesized ECG value and N denotes the total number of data points. SMAPE returns a scale-independent relative error.

IV. Results

A. *Quantitative ECG Analysis*

DMTT is evaluated quantitatively using two metrics: Average Absolute Error (AAE) and Normalized Root Mean Squared Error (NRMSE). Average Absolute Error quantifies the average absolute difference between the synthesized and real ECGs. AAE, measured in millivolts, allows direct comparison of synthesized ECGs to original signals, assessing the model's ability to replicate disease-specific and normal patterns. A low AAE means the ECG is capturing the underlying data patterns accurately.

$$AAE = \frac{\sum_{i=i}^{N}|s_{pred,i}-s_{true,i}|}{N} \qquad (17)$$

NRMSE is a scaled metric that does not directly correspond to the input units. NRMSE is the difference between the synthesized ECG and actual ECG, while accounting for the data scale through normalization, as shown in (18). For ECG synthesis, NRMSE is a useful metric for comparing the accuracies across different leads, conditions, and datasets, as ECGs can have different scales. NRMSE ensures that errors are not inflated or underestimated due to natural amplitude variation and different recoding systems.

$$NRMSE = \frac{\sqrt{\sum_{i=1}^{N}\frac{\left(s_{pred,i}-s_{true,i}\right)^2}{N}}}{s_{pred,max}-s_{pred,min}} \qquad (18)$$

Table 2. AAE across V1-V5 for the five disease types

Leads	AAE of Synthesized Leads V1-V5					
	Normal	ASMI	LAFB	LVH	IMI	Average
Lead V1	0.0425	0.0670	0.0546	0.0643	0.0495	0.0556
Lead V2	0.0769	0.1232	0.1030	0.1228	0.0919	0.1036
Lead V3	0.0759	0.1278	0.1000	0.1232	0.0906	0.1035
Lead V4	0.0524	0.1013	0.0738	0.1010	0.0714	0.0800
Lead V5	0.0317	0.0546	0.0454	0.0572	0.0401	0.0458

Table 3. NRMSE across V1-V5 for the five disease types

Leads	NRMSE of Synthesized Leads V1-V5					
	Normal	ASMI	LAFB	LVH	IMI	Average
Lead V1	0.0798	0.0956	0.1145	0.0724	0.0940	0.0913
Lead V2	0.0939	0.1402	0.1265	0.1078	0.1082	0.1153
Lead V3	0.0880	0.1483	0.0992	0.1036	0.1007	0.1079
Lead V4	0.0541	0.1462	0.0729	0.0777	0.0802	0.0862
Lead V5	0.0345	0.0852	0.0516	0.0443	0.0476	0.05264

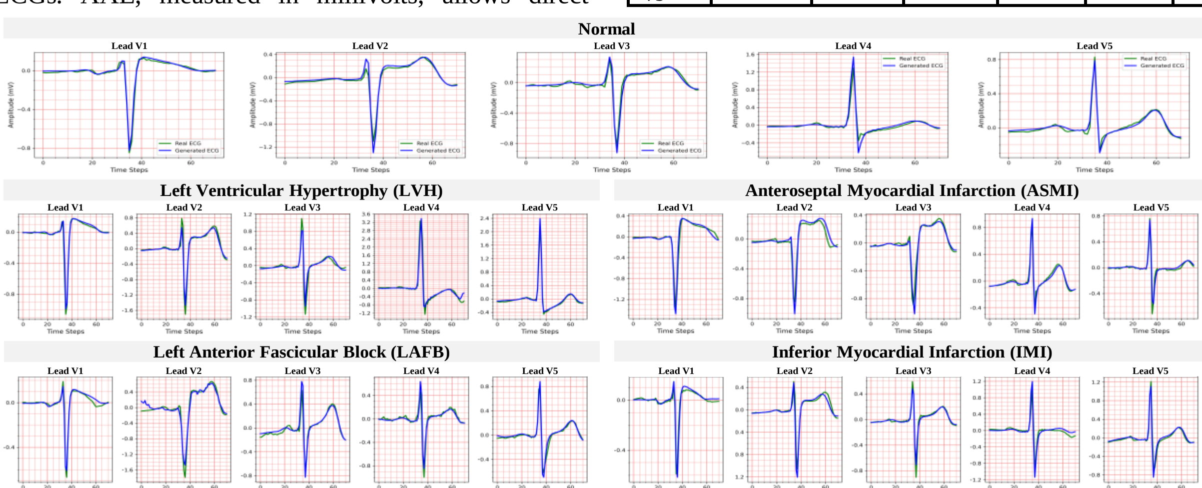

Figure 6. Qualitative results showing synthesized (blue) and actual (green) ECG waveforms for leads V1-V5, across the five disease types

B. *Qualitative ECG Analysis*

Since ECGs are interpreted by human clinicians, it is crucial to ensure that the synthesized ECGs are visually faithful to the original. Fig. 6 shows the synthesized and actual ECGs waves superimposed to illustrate their degree of similarity. The images show that the key morphological attributes of the P-waves, QRS complexes, T-waves are reproduced accurately for ECGs of all types, i.e., normal and diseased and across all five leads (V1-V5).

V. Previous Work

Prior studies have employed various generative AI techniques to synthesize ECGs, depending on the specific goals and applications. These include Convolutional Neural Network, Long Short-Term Memory Neural Networks, and Generative Adversarial Networks to generate missing leads with varying degrees of success [14], [15], [16], [17]. Some of these approaches also leverage transfer learning to improve patient-specific performance, while others focus on generalizing ECG synthesis across diverse datasets. The proposed approach differentiates itself through the development of a systematic method of analyzing the ECG characteristics using a specialized transformer implementation. This model is specifically trained to identify the distinct attributes of normal and abnormal ECGs, thereby achieving a comprehensive feature extraction, resulting in a high accuracy across multiple disease types, making it a promising approach to ECG synthesis.

VI. Conclusion

This study explored the effectiveness of the Dynamic Multi-scale Timeseries Transformer (DMTT) for ECG synthesis using only three leads as input (I, III, and V6). ECGs from five disease types were analyzed to demonstrate the model's applicability to both normal ECGs and those exhibiting various cardiovascular diseases. The quantitative analysis indicates a low AAE and RMSE, suggesting high accuracy. Qualitative analysis of the results, presented in Fig. 6, indicates that the synthesized ECGs closely replicate the morphology and timing of actual ECGs across various cardiovascular conditions. While certain leads could benefit from further refinement, the overall novel framework demonstrates the viability of the proposed technique in generating clinically meaningful ECGs.

Future research will incorporate individual biometrics (e.g., age, gender, race, weight) to personalize ECG synthesis, ensuring generated signals reflect individually tailored cardiac profiles and pre-existing conditions.

References

[1] GY. Li, G. Y. Cao, W. Z. Jing, J. Liu, and M. Liu, “Global trends and regional differences in incidence and mortality of cardiovascular disease, 1990–2019: Findings from 2019 global burden of disease study,” Eur. J. Prev. Cardiol., vol. 30, no. 3, pp. 276–286, Feb. 2023.

[2] C. E. Kossmann et al., “Recommendations for standardization of leads and of specifications for instruments in electrocardiography and vectorcardiography,” Circulation, vol. 35, no. 3, pp. 583–602, 1967.

[3] W. Einthoven, “The telecardiogram,” Am. Heart J., vol. 53, no. 4, pp. 602–615, 1957.

[4] E. Goldberger, “A simple, indifferent, electrocardiographic electrode of zero potential and a technique of obtaining augmented, unipolar, extremity leads,” Am. Heart J., vol. 23, no. 4, pp. 483–492, 1942.

[5] E. Goldberger, “The aVL, aVR, and aVF leads: A simplification of standard lead electrocardiography,” Am. Heart J., vol. 24, no. 3, pp. 378–396, 1942.

[6] A. R. Barnes et al., “Standardization of precordial leads: Supplementary report,” Am. Heart J., vol. 15, no. 2, pp. 235–239, 1938.

[7] C. Nataraj, A. Jalali, and P. Ghorbanian, Application of Computational Intelligence Techniques for Cardiovascular Diagnostics. New York, NY, USA: Springer, 2012.

[8] StoryMD. “ECG Abnormalities.” StoryMD Journal. Feb. 8, 2025. [Online]. Available: https://storymd.com/journal/joy3e7lsnw-heart-s-electrical-activity/page/5dzlpi96egd-ecg-abnormalities.

[9] A. Samol et al., “Patient directed recording of a bipolar three-lead electrocardiogram using a smartwatch with ECG function,” J. Vis. Exp., no. 154, Dec. 2019.

[10] J. van der Zande et al., “Using a smartwatch to record precordial electrocardiograms: A validation study,” Sensors (Basel), vol. 23, no. 5, p. 2555, Feb. 2023.

[11] P. Wagner et al., “PTB-XL, a large publicly available electrocardiography dataset,” Sci. Data, vol. 7, no. 1, p. 154, May 2020.

[12] A. Sommers et al., “A survey of transformer enabled time series synthesis,” arXiv preprint arXiv:2406.02322, 2024.

[13] A. Vaswani et al., “Attention is all you need,” arXiv preprint arXiv:1706.03762, 2017.

[14] A. Malpani, J. Xue, D. Albert, and G. Clifford, “A transfer learning model-based patient-specific ECG lead synthesis algorithm,” in Proc. Comput. Cardiol., 2024, pp. 1–4.

[15] M. Matyschik, H. Mauranen, P. Bonizzi, and J. Karel, “Feasibility of ECG reconstruction from minimal lead sets using convolutional neural networks,” in Proc. 2020 Comput. Cardiol., Rimini, Italy, 2020, pp. 1–4.

[16] J. Sohn, S. Yang, J. Lee, Y. Ku, and H. C. Kim, “Reconstruction of 12-lead electrocardiogram from a three-lead patch-type device using a LSTM network,” Sensors, vol. 20, p. 3278, 2020.

[17] Y.-H. Zhang and S. Babaeizadeh, “Synthesis of standard 12-lead electrocardiograms using two-dimensional generative adversarial networks,” J. Electrocardiol., vol. 69, pp. 6–14, 2021.

[18] P. Ren et al., “Beyond fixation: Dynamic window visual transformer,” arXiv preprint arXiv:2203.12856, 2022.

[19] S. Chen et al., “DWFormer: Dynamic window transformer for speech emotion recognition,” arXiv preprint arXiv:2303.01694, 2023.

[20] H. Fan et al., “Multiscale vision transformers,” arXiv preprint arXiv:2104.11227, 2021.